A Copula-Guided In-Model Interpretable Neural Network for Change Detection in Heterogeneous Remote Sensing Images

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***Abstract*— Change detection (CD) in heterogeneous remote sensing images has been widely used for disaster monitoring and land-use management. In the past decade, the heterogeneous CD problem has significantly benefited from the development of deep neural networks (DNNs). However, the purely data-driven DNNs perform like a black box where the lack of interpretability limits the trustworthiness and controllability of DNNs in most practical CD applications. As a powerful knowledge-driven tool, copula theory performs well in modeling dependence among random variables. To enhance the interpretability of existing neural networks for heterogeneous CD, we propose a knowledge-data-driven heterogeneous CD method based on a copula-guided neural network, named NN-Copula-CD. In our NN-Copula-CD, the mathematical characteristics of copula are employed as the loss functions to supervise a neural network to learn the dependence between bi-temporal heterogeneous superpixel pairs, and then the changed regions are identified via binary classification based on the degrees of dependence of all the superpixel pairs in the bi-temporal images. We conduct in-depth experiments on four datasets with heterogeneous images, including synthetic aperture radar (SAR), multispectral, and near-infrared images, where quantitative and visual results demonstrate the effectiveness and interpretability of our proposed NN-Copula-CD method.**

***Index Terms*— Change detection (CD), copula theory, heterogeneous remote sensing images, neural networks, remote sensing.**

I. Introduction

CHANGE detection (CD) is an important technology that has been applied to a large number of Earth observation tasks [2], [3], such as disaster monitoring [4], [5], land-use management [6], [7], and forest protection [8]. The CD serves to monitor natural and man-made changes in the same area by analyzing differences among multitemporal remote sensing images [9]. Depending on the characteristics of remote sensing images involved in CD, the CD tasks can be categorized into homogeneous and heterogeneous CD tasks [10]. The homogeneous CD task relies on remote sensing images captured by the same sensors and has been extensively investigated during the past decades [10]. However, the satellites' revisit cycles and weather conditions often limit the timeliness of homogeneous CD in urgent events [11], [12]. With the proliferation of satellites equipped with various imaging sensors in recent years, heterogeneous remote sensing images can be more rapidly accessed and offer more real-time information on land-cover objects [11], [13]. Thus, heterogeneous CD has shown its great capabilities and utilities in practical CD applications, especially for time-sensitive emergencies [4], [12], and has been attracting more and more attention from scholars and practitioners.

Due to the different characteristics of remote sensing images acquired by various sensors, the classic algebraic calculation in [14], [15], and [16] is not feasible for heterogeneous CD. To solve this problem, Wan et al. [17] exploit compound classification to classify the heterogeneous images with sets of categories and then perform decision-level fusion based on the classification results to determine the location of changes. The postclassification treatment in [17] for heterogeneous images suppresses the false alarms/missed detection caused by the impact of heterogeneity to a certain extent, yet its performance relies heavily on the abilities of classification algorithms employed for heterogeneous images.

It is also a widely adopted approach that transforms bi-temporal images into the same representation space where they are easier to be compared [13]. Following this idea, an earlier study [18] utilizes the 3-D information of the land-cover objects and the imaging parameters of the postchange image to predict the characteristics of the land-cover objects in the representation space of the postchange image (assuming no change occurs). Then, the similarity between the predicted characteristics and the real characteristics in the postchange scene is calculated to identify if any changes have occurred. Since the satellite parameters are difficult to acquire in most cases, the later study [19] estimates the transformation functions directly from remote sensing images via semi-supervised training based on the K-nearest neighbor (KNN) algorithm. The transformation method in [19] achieves superior performance than [18] in

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situations when less *a priori* information (e.g., imaging parameters) is available. However, the computational cost of training in [19] increases significantly when searching for the KNNs in a massive number of training samples from large remote sensing images [4], [19].

In recent years, many powerful heterogeneous CD methods have been proposed due to the rapid advancement and strong representation capabilities of deep neural networks (DNNs) [10]. In [4] and [20], deep convolution networks with symmetric structures are employed to transform bi-temporal heterogeneous images into a common space for comparisons. Jiang et al. [21] employ deep image style transfer networks to convert bi-temporal heterogeneous images into a homogeneous style space, preserving the semantic contents of each image and transforming the textures of bi-temporal images into a uniform style. In [22], [23], and [24], generative adversarial networks (GANs) are utilized to translate one image into the domain of the other image, which significantly reduces the heterogeneity between images of different modalities and enhances the saliency of changes. In addition, the autoencoders in [25] and [26] align the features of bi-temporal images of different modalities in a latent space via adversarial training to facilitate the learning of the image-to-image translation functions. Although the aforementioned DNN-based methods have made significant progress for the heterogeneous CD task, the operating mechanisms and internal principles of DNNs are not available to us in most cases, resulting in a lack of interpretability[1] in practical CD applications.

Copula theory describes the joint cumulative distribution function (cdf) of multivariate random variables via a copula function and their respective marginal cdfs and is an effective tool to model statistical dependence among random variables [27]. Meanwhile, it has been demonstrated in the pioneering works [28], [29] that accurate statistical dependence modeling of bi-temporal data plays a key role in enhancing CD performance. Thus, the copula theory is highly appropriate for the CD task. In [29], copula theory is applied to capture the dependence between bi-temporal heterogeneous images in unchanged areas, and then the copula-based regression is performed on the prechange image to calculate the simulated statistics at the postchange time. Finally, the changed information is measured by comparing the simulated statistics with the real statistics in the postchange image. Note that the CD performance could be seriously degraded in case an inappropriate copula function is selected. The copula functions used in [29] are manually selected from the existing copula families based on the visual inspection of the dependence of the selected samples. In practice, the dependence in heterogeneous image pairs varies significantly with different scenarios and sensors. Thus, selecting a suitable copula function from the existing copula families for various scenes is labor-consuming. Besides, many existing copula functions are customized for other fields such as finance [30], which limits the usability of copula theory in remote sensing tasks.

In recent years, some studies [31], [32], [33] have attempted to model copula functions via neural networks. Thanks to the powerful and flexible representation capabilities of neural networks, the copula functions learned by neural networks exhibit a comparable or even superior performance than the traditional copula functions. Meanwhile, the intrinsic essence of CD has been demonstrated to have a close relationship with copula theory [28], [29], which naturally provides an opportunity to introduce copula theory into neural networks for CD. Inspired by the insights in [31], [32], and [33], we propose a novel semi-supervised heterogeneous CD method based on a copula-guided neural network, named NN-Copula-CD, to obtain better interpretability than the existing DNN-based heterogeneous CD methods while enhancing the generalization capabilities of traditional copula functions applied for remote sensing data. The core idea of our NN-Copula-CD is to estimate the degrees of dependence between registered regions in bi-temporal images via a copula-guided neural network. First, our NN-Copula-CD trains a neural network to learn a copula function from a few unchanged superpixel[2] pairs. The trained copula function represents the unchanged patterns where the unchanged superpixel pairs tend to be strongly dependent and the changed superpixel pairs tend to be weakly dependent. Next, we exploit the trained copula function to calculate the degrees of dependence for all superpixel pairs in the bi-temporal images, and then the changed and unchanged superpixel pairs can be easily distinguished via the degrees of dependence. Note that compared with most existing heterogeneous CD methods based on DNNs, the neural network in our NN-Copula-CD has better interpretability based on copula theory, that is, mathematical constraints that a copula function should satisfy are employed as loss functions to guide the training process of the neural network. This enables the trained neural network to follow the mathematical principles of copula theory and approximate a copula function that accurately reflects the dependence of unchanged regions in image pairs.

The main contribution of our work is that a new knowledge-data-driven CD method based on the copula-guided neural network, that is, NN-Copula-CD, is proposed to tackle the heterogeneous CD problem. In our NN-Copula-CD, a neural network is trained under the guidance of copula theory. Then, we exploit it to model the dependence between the prechange and postchange superpixels to identify the changed areas. During this process, the generalization capabilities of data-driven neural networks are integrated with the theoretical basis of knowledge-driven copula theory, which improves the interpretability of the DNN-based heterogeneous CD methods and the generalization capabilities of the traditional-copula-based heterogeneous CD methods. Compared to our conference version [1], we add more comparative experiments, datasets, visualization analysis, discussion, and implementation details of the proposed NN-Copula-CD. Experimental results on four heterogeneous CD datasets demonstrate the superiority of the proposed NN-Copula-CD in terms of the kappa coefficient. Visual results also exhibit the effectiveness and interpretability of our method.

[1]We will give the definition and taxonomy of interpretability in Section II-B.

[2]The notion of superpixels will be defined later in Section III-B.

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The rest of this article is organized as follows. Section II reviews copula theory, the definition and taxonomy of interpretability, and the existing heterogeneous CD methods based on neural networks. Section III presents the formulation and specifics of the proposed NN-Copula-CD method. Section IV provides the in-depth experiments and discussion on four heterogeneous CD datasets. Finally, the conclusion and our future directions are given in Section V.

II. Background and Related Works

In this section, a brief background on copula theory and interpretability along with a more complete discussion of the literature that is directly related to our work are presented.

A. Copula Theory

Sklar's theorem [34] indicates that the joint distribution of multivariate random variables can be represented in terms of all the marginal cdfs of random variables and a copula function. Let $F(v_1, v_2, \ldots, v_d)$ denote the joint cdf of random variables, and their marginal cdfs are represented by $F_1(v_1), F_2(v_2), \ldots, F_d(v_d)$, respectively. There always exists a d-dimensional copula function C, where

$$F(v_1, v_2, \ldots, v_d) = C(F_1(v_1), F_2(v_2), \ldots, F_d(v_d)). \quad (1)$$

By utilizing an appropriate copula function, Sklar's theorem allows one to connect the dependence between random variables with the marginal distributions of each random variable [35], which provides a convenient and effective way to deal with complex dependence modeling problems in finance [30], signal processing [36], and remote sensing [28], [29].

Copula theory has been applied for remote sensing applications in some prior works. Wang et al. [37] proposed a copula-based fusion strategy that utilizes clutter correspondence between multisource synthetic aperture radar (SAR) images to enhance the target detection performance. Mercier et al. [29] proposed a CD model that utilizes conditional copulas to generate the simulated statistics from prechange statistics. These simulated statistics are then compared with the real statistics in the postchange image via a modified Kullback–Leibler divergence to measure the changes. Note that the copula functions used in [29] need to be specified by visual inspection of dependence. Li et al. [28] replaced the visual inspection process of selecting copula functions in [29] with an analytical copula selection strategy. However, since many traditional copula functions are still custom-made for finance and other fields, it is difficult to find a generalized copula function for [28] to fit remote sensing data of different scenarios.

B. Interpretability of ML Methods

With the development of machine learning (ML) methods (especially neural networks) in various fields, the interpretability of data-driven ML methods has received increasing attention and discussion from scholars [38], [39], [40]. A general definition for interpretability is the capability that offers explanations in understandable terms to make humans interpret the decision-making mechanism of ML methods [39], [40]. Just like humans can explain the same thing from different perspectives, interpretable ML methods can also be divided into various taxonomies with different criteria [39], [41]. Here, we introduce the taxonomy of interpretable ML methods used in [39] and [41], where interpretability in ML methods is classified into three categories, that is, pre-model interpretability, in-model interpretability, and post-model interpretability.

The pre-model interpretability usually comes from the interpretable operations that analyze data before model building and training [41]. For example, exploratory data analysis methods such as data visualization can analyze correlations and regularity in data and features to understand the patterns of inputs for models [39]. The in-model interpretability represents the inherent interpretability within the models, which can be implemented by the theoretical rules and constraints used in building and training models [39], [41]. For example, linear regression (LR) is a typical ML model with in-model interpretability of logical rules and mathematical constraints. Additionally, neural networks can incorporate in-model interpretability. A notable example is the physical information neural network (PINN) [42], which integrates theoretical rules and mathematical constraints into the training of neural networks to approximate certain physical regulations. These theoretical rules and mathematical constraints are often implemented as different loss functions to guide the training of neural networks, where the decision-making process of the trained neural networks is potentially supervised by the corresponding physical regulations. Post-model interpretability mainly emerges after the models have been trained [39]. One of the most well-known examples of post-model interpretability is the gradient-weighted class activation mapping (Grad-CAM) method [43]. Meanwhile, post-model interpretability can also be achieved by mimic models, which train an interpretable model to approach the predictions of an uninterpretable trained model [39].

Note that in this article, we consider the interpretability in the proposed NN-Copula-CD as in-model interpretability because the neural copula model in NN-Copula-CD is trained under the guidance of copula theory, which is similar to the idea of PINN [42].

C. Neural Networks for Heterogeneous CD

In the past decade, neural networks have been widely used for various remote sensing tasks [44], [45], [46] due to their powerful nonlinear modeling capabilities. In recent CD studies, image/feature domain alignment and dependence modeling have been widely investigated using neural networks for both homogeneous and heterogeneous CD. In a recent homogeneous CD method [47], a novel deep network based on attention mechanisms is proposed to achieve bi-temporal feature alignment in temporal and multiscale levels. Chen et al. [48] designed a scale-invariant deep network to address the cross-resolution CD problem, where a local window self-attention module is applied to enhance feature alignment of remote sensing images with different resolutions. Meanwhile, some transformer-based networks have achieved

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impressive performance in terms of dependence modeling. The transformers in [49] and [50] exploit attention mechanisms in DNNs to facilitate modeling long-range spatial dependence and temporal dependence of bi-temporal images.

Regarding heterogeneous CD tasks, various neural networks including image-to-image translation networks and deep homogeneous transformation networks have been utilized to achieve domain alignment and address the challenge of heterogeneity. Image-to-image translation networks translate images from the domain of one temporal image into the domain of the other temporal image while maintaining the semantic content [51]. Deep homogeneous transformation networks simultaneously transform images into a common deep feature space where the feature representations of two images are approximately homogeneous [20].

For the image-to-image translation networks [22], [23], [24], [25], Niu et al. [24] employed the conditional GAN (cGAN) to translate one image into the representation space of the other image and then directly compared the translated source image and the original target image to obtain the CD results. Liu et al. [23] and Luppino et al. [25] extended Niu's work [24] by improving the cGAN with the idea of cycle-consistent adversarial networks (CycleGANs) [52] that provides superior stability and translation performance. The noise in the difference map between the translated prechange image and the original postchange image is also considered in [23], where a random forest classifier replaces direct comparison to improve the robustness of CD results. In addition, Jiang et al. [21] leveraged the different layers of the deep convolutional network to extract semantic features and style features and then transferred the style features of bi-temporal images into the homogeneous space while keeping semantic features. Then, the changed areas can be easily recognized via a homogeneous CD method.

For the deep homogeneous transformation networks, the coupling network in [20] simultaneously transforms the bi-temporal heterogeneous images into the deep homogeneous feature space via deep coupling convolution layers, where the extracted deep homogeneous features can be used to calculate distance maps and identify the changes. Furthermore, a Siamese network in [4] focuses on deep-level feature transformation, and the efficiency of CD is improved by saving network parameters that are required to be optimized.

Note that the aforementioned neural networks [4], [20], [21], [22], [23], [24], [25] have delivered outstanding performance for the heterogeneous CD task, the lack of interpretability therein is still an issue. This deficiency results in limited controllability since the operating mechanisms of most existing neural networks for heterogeneous CD work as a black box for us. It is also noticed that probabilistic models have recently been utilized in a neural networks-based CD method [53], where a bipartite neural network based on probabilistic models is trained via an energy function to describe the joint distribution of bi-temporal images. Note that, compared with [53], our NN-Copula-CD focuses on establishing the joint distribution of unchanged areas so that the saliency of changed areas can be relatively emphasized.

III. Proposed NN-Copula-CD Method

In this section, our proposed NN-Copula-CD method is introduced. We begin by formulating the heterogeneous CD problem and presenting the overall framework of the proposed method. Then, we introduce the notions of superpixel segmentation and training sample selection, the training of the copula-guided interpretable neural network for heterogeneous CD, and the generation of final detection results in our NN-Copula-CD method.

A. Problem Formulation and Overview of the Approach

Assume that I^{T_1} and I^{T_2} are bi-temporal co-registered heterogeneous remote sensing images of the same location at the time T_1 and the time T_2, respectively. The task is to recognize the changes between I^{T_1} and I^{T_2} and produce pixel-level binary results to indicate the locations of the changes.

We first present a high-level overview of the proposed NN-Copula-CD method as well as definitions in this section. We will provide more details later. In the first step, we segment the bi-temporal images into superpixel pairs and select a few unchanged superpixel pairs as the training samples. Then, a fully connected neural network is trained under the guidance of copula theory to learn the dependence between the unchanged superpixel pairs. Next, the trained neural copula is exploited to calculate the degrees of dependence for all the superpixel pairs in the bi-temporal images, where the degrees of dependence are high (low) in the unchanged (changed) regions. Finally, the fuzzy c-means algorithm [54] is utilized to classify all the superpixel pairs into the changed and unchanged categories according to the calculated degrees of dependence.

The overall framework of the proposed method is shown in Fig. 1. Our method is semi-supervised and the training process can be mainly divided into three steps (see the blue stream in Fig. 1), that is, superpixel segmentation, training sample selection, and neural copula training. The test process mainly consists of three steps (see the red stream in Fig. 1), that is, superpixel segmentation, neural copula inference, and clustering and classification using the fuzzy c-means algorithm.

B. Superpixel Segmentation and Training Sample Selection

In most remote sensing applications, the acquired images are quite large. In addition, heterogeneity caused by different sensors also poses serious challenges for the pixel-level dependence analysis. Therefore, it may not be efficient to directly build a pixel-to-pixel dependence model for high-resolution heterogeneous remote sensing images. Compared with individual pixels, superpixels are the segmentation results of perceptual clustering algorithms that group adjacent pixels with similar visual characteristics [55]. Consequently, superpixels provide a more comprehensive representation of a group of pixels and are commonly used as basic units for heterogeneous CD analysis [11], [12], [13], [56]. In the proposed NN-Copula-CD method, the superpixel segmentation method in [13] based on the simple linear iterative clustering (SLIC) algorithm [55] is applied to segment bi-temporal heterogeneous images into superpixel pairs. The main steps of the

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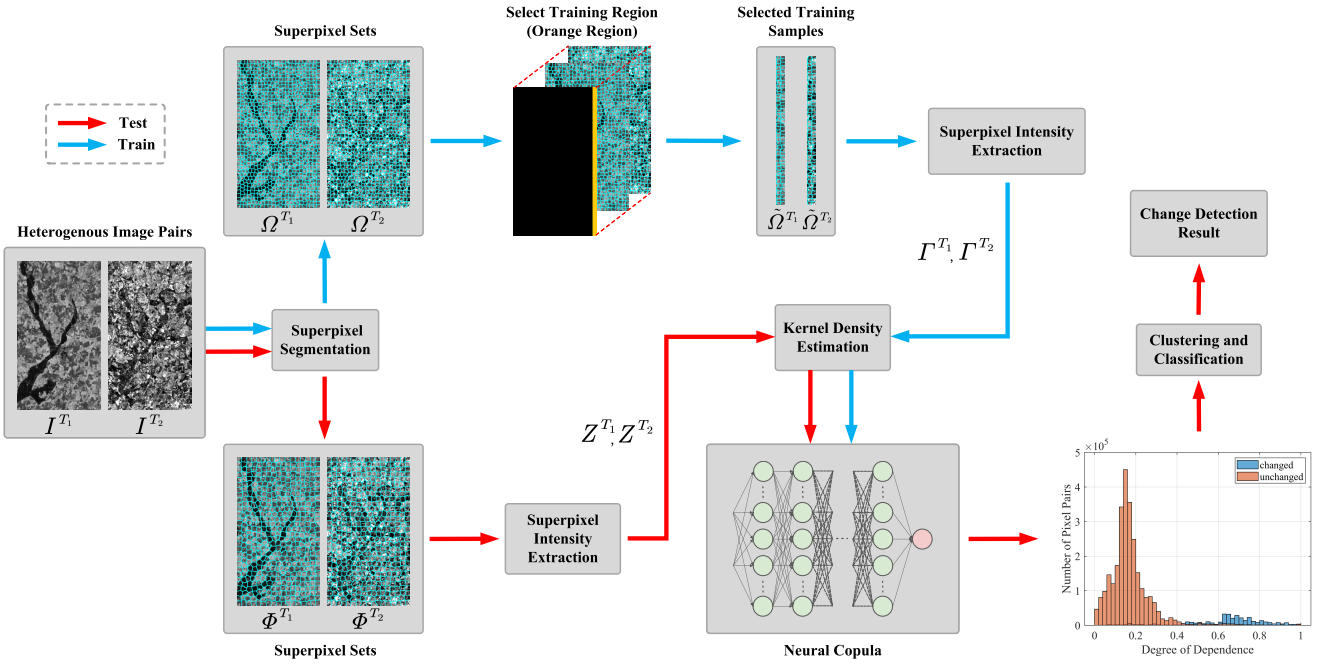


Fig. 1. Overall framework of the proposed NN-Copula-CD.

superpixel segmentation method [13] are as follows: First, the SLIC method is applied on each heterogeneous image independently to obtain two superpixel segmentation maps. Then, the obtained two superpixel segmentation maps are merged into one superpixel segmentation map. Finally, the bi-temporal heterogeneous images share the same superpixel segmentation map. The superpixel sets of I^{T_1} and I^{T_2} after the superpixel segmentation method [13] are denoted by Ω^{T_1} and Ω^{T_2}, respectively, and we use N_1 to denote the number of segmented superpixels in both I^{T_1} and I^{T_2}. $\Omega_i^{T_1}$ and $\Omega_i^{T_2}$ are the aligned ith superpixels in Ω^{T_1} and Ω^{T_2}, respectively.

Once the superpixel segmentation is finished, the next step is to train the neural copula model. Due to the inconsistent distributions of heterogeneous data across different scenes, training a neural copula model offline from other scenes using historical data often leads to inferior CD performance. In light of this, we choose a few unchanged regions from the original images to train the neural copula model online, like the existing heterogeneous CD methods [4], [19]. However, the training regions are manually picked by professionals in [4] and [19] and the training sample selection process therein is time-consuming and labor-intensive. One of the reasonable strategies is to choose the regions near the edge of satellite images as training regions. In most emergencies, satellites will ensure that the coverage of the captured images is wide enough to fully cover the area of change that is of interest (such as the epicenter) [57], which means that the captured images usually contain unchanged areas at the edge of the large satellite images. In this work, we follow [28] to select a region nearby the edge of the input bi-temporal images as the selected training region (see the orange region as shown in Fig. 1). This strategy also saves time and labor costs in selecting training samples. We will further discuss the impact on performance caused by the selection of training samples in Section IV. Superpixels will be picked from Ω^{T_1} and Ω^{T_2} for training based on whether the number of intersecting pixels between the current superpixel and the selected training region is larger than half of the number of pixels within the current superpixel. Suppose that Λ is the selected training region, the process of superpixel selection for training is expressed as

$$\tilde{\Omega}^{T_1} = \left\{ \Omega_j^{T_1} | \frac{\mathrm{Num}\left(\Omega_j^{T_1} \cap \Lambda\right)}{\mathrm{Num}\left(\Omega_j^{T_1}\right)} > 0.5, \Omega_j^{T_1} \in \Omega^{T_1} \right\}$$
$$\tilde{\Omega}^{T_2} = \left\{ \Omega_j^{T_2} | \frac{\mathrm{Num}\left(\Omega_j^{T_2} \cap \Lambda\right)}{\mathrm{Num}\left(\Omega_j^{T_2}\right)} > 0.5, \Omega_j^{T_2} \in \Omega^{T_2} \right\} \quad (2)$$

where $\mathrm{Num}(\cdot)$ is the function that counts the number of pixels. $\cap$ denotes the intersection operation between regions. $\tilde{\Omega}^{T_1}$ and $\tilde{\Omega}^{T_2}$ are the selected training superpixel sets both with N_2 superpixels from I^{T_1} and I^{T_2}, respectively. We use the mean value of all the pixels within a training superpixel, that is, superpixel intensity, to concisely represent that superpixel, that is,

$$\Gamma^{T_1} = \left\{ \gamma_k^{T_1} = \frac{\mathrm{Floor}\left(\mathrm{Mean}\left(\tilde{\Omega}_k^{T_1}\right)\right)}{255}, 1 \le k \le N_2 \right\}$$
$$\Gamma^{T_2} = \left\{ \gamma_k^{T_2} = \frac{\mathrm{Floor}\left(\mathrm{Mean}\left(\tilde{\Omega}_k^{T_2}\right)\right)}{255}, 1 \le k \le N_2 \right\} \quad (3)$$

where $\mathrm{Mean}(\cdot)$ is the function that calculates the mean value of all the pixels in the corresponding superpixel, and $\mathrm{Floor}(\cdot)$ is used to truncate the decimal parts of numbers. Since the range of pixel values in most remote sensing images is 0 to 255, we normalize the superpixel intensity via a division of 255. Γ^{T_1} and Γ^{T_2} are the sets containing the normalized

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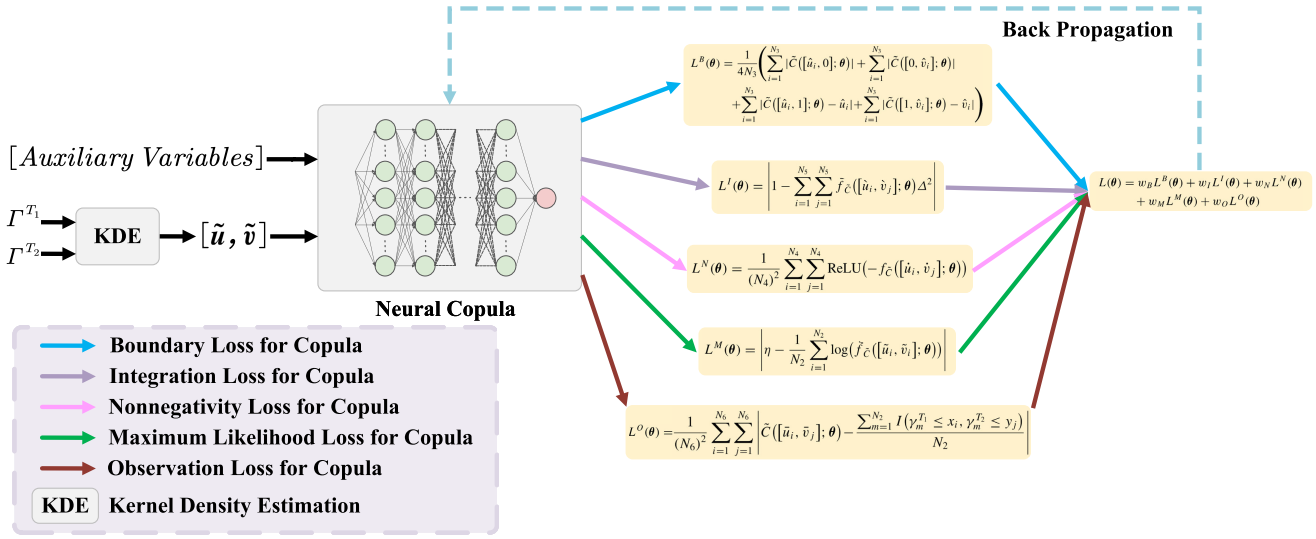


Fig. 2. Training diagram of the neural copula in our NN-Copula-CD. The auxiliary variables contain the vectors $[\hat{\boldsymbol{u}}, \hat{\boldsymbol{v}}]$, $[\grave{\boldsymbol{u}}, \grave{\boldsymbol{v}}]$, $[\dot{\boldsymbol{u}}, \dot{\boldsymbol{v}}]$, and $[\bar{\boldsymbol{u}}, \bar{\boldsymbol{v}}]$ that facilitate the neural copula model to satisfy the boundary constraint, integration constraint, nonnegativity constraint, and observation constraint, respectively.

superpixel intensity of the training superpixels in $\tilde{\Omega}^{T_1}$ and $\tilde{\Omega}^{T_2}$, respectively. It is worth mentioning that there might be more features (e.g., variance and median value) available to represent the characteristics of superpixels. In this article, we only utilize the mean values to show the effectiveness of our copula-guided CD method because our focus is not to exhaustively exploit the features of superpixels. We refer the readers to [12] and [13] for more superpixel features used in the CD task.

C. Neural Copula Model

Once the normalized superpixel intensity sets Γ^{T_1} and Γ^{T_2} are obtained, the next step is to model the dependence relationship between Γ^{T_1} and Γ^{T_2} via copula functions. Traditional copula functions are utilized based on *a priori* knowledge of data distributions, focusing on the specific dependence relationships of interest [58]. For example, Clayton copula focuses on lower tail dependence while Frank copula is concerned about symmetric tail dependence. However, *a priori* knowledge of the distribution of heterogeneous remote sensing data is difficult to know precisely due to various imaging conditions and scenarios. Therefore, we are often uncertain about which traditional copula function is the most appropriate in advance. To address this issue, we adopt a neural copula model in [33] to learn the dependence between unchanged regions from remote sensing data.

The first step of copula theory is to apply the probability integral transform for each marginal distribution, that is, to estimate the marginal cdf values for every element in Γ^{T_1} and Γ^{T_2}. In our implementation, we utilize the kernel density estimation (KDE) approach [59] to estimate the marginal cdf values. Note that the original KDE is designed for probability density function (pdf) estimation, and we calculate integration on the estimated pdf to obtain the cdf values. Suppose that $\mathrm{KDE}(R_1; R_2)$ denotes the KDE function that estimates the marginal cdf values at elements in the set R_1 according to the distribution fit by elements in the set R_2. The estimation for marginal cdf values of Γ^{T_1} and Γ^{T_2} can be formulated as

$$\begin{aligned} \tilde{\boldsymbol{u}} &= \mathrm{KDE}\left(\Gamma^{T_1}; \Gamma^{T_1}\right) \\ \tilde{\boldsymbol{v}} &= \mathrm{KDE}\left(\Gamma^{T_2}; \Gamma^{T_2}\right) \end{aligned} \tag{4}$$

where $\tilde{\boldsymbol{u}} = [\tilde{u}_1, \tilde{u}_2, \ldots, \tilde{u}_{N_2}]$ and $\tilde{\boldsymbol{v}} = [\tilde{v}_1, \tilde{v}_2, \ldots, \tilde{v}_{N_2}]$ are the marginal cdf values estimated from Γ^{T_1} and Γ^{T_2}, respectively. Since the normalized superpixel intensities in Γ^{T_1} and Γ^{T_2} range from 0 to 1 with 256 evenly spaced points, we exploit KDE to estimate the marginal cdf values at these evenly spaced points and then construct cdf value tables. Next, we obtain the marginal cdf values for every element in Γ^{T_1} and Γ^{T_2} by table lookup of the established cdf value tables. This lookup table operation can significantly alleviate duplicate computation in the marginal cdf estimation process.

Next, we introduce the network structure and the training process of the neural copula. As shown in Fig. 2, the neural copula model is a fully connected neural network that includes three parts, that is, the input layer, the hidden layers, and the output layer. In our implementation, we construct a fully connected neural network with five hidden layers and twenty neurons per hidden layer. As in [33], the activation functions for hidden layers and output layer are *tanh* and *sigmoid* functions, respectively. Suppose that the network is written as $\tilde{C}([\boldsymbol{u}, \boldsymbol{v}]; \boldsymbol{\theta})$, where $\boldsymbol{\theta}$ denotes the trainable parameters. The inputs of the network are $[\boldsymbol{u}, \boldsymbol{v}]$. The outputs of the network provide the values of the copula function and the gradient $f_{\tilde{C}}([\boldsymbol{u}, \boldsymbol{v}]; \boldsymbol{\theta})$ of the network denoting the copula pdf. Formally,

$$f_{\tilde{C}}([\boldsymbol{u}, \boldsymbol{v}]; \boldsymbol{\theta}) = \frac{\partial^2 \tilde{C}([\boldsymbol{u}, \boldsymbol{v}]; \boldsymbol{\theta})}{\partial \boldsymbol{u} \partial \boldsymbol{v}}. \tag{5}$$

Afterward, we tap the potential of the neural network for the CD task by training it under the guidance of copula theory. Inspired by [33], the fully connected neural network converges into a copula function based on five mathematical loss functions.

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1) Boundary Constraint of Neural Copula: The outputs of the network should obey the boundary constraint of copula's definition, that is,

$$\begin{cases} \tilde{C}([\boldsymbol{u},\boldsymbol{0}];\boldsymbol{\theta}) = \boldsymbol{0} \\ \tilde{C}([\boldsymbol{0},\boldsymbol{v}];\boldsymbol{\theta}) = \boldsymbol{0} \\ \tilde{C}([\boldsymbol{u},\boldsymbol{1}];\boldsymbol{\theta}) = \boldsymbol{u} \\ \tilde{C}([\boldsymbol{1},\boldsymbol{v}];\boldsymbol{\theta}) = \boldsymbol{v}. \end{cases} \tag{6}$$

$[\boldsymbol{u},\boldsymbol{0}]$, $[\boldsymbol{u},\boldsymbol{1}]$, $[\boldsymbol{0},\boldsymbol{v}]$, and $[\boldsymbol{1},\boldsymbol{v}]$ represent the boundary points of the copula function. Suppose that $\hat{\boldsymbol{u}} = [\hat{u}_1, \hat{u}_2, \ldots, \hat{u}_{N_3}]$ and $\hat{\boldsymbol{v}} = [\hat{v}_1, \hat{v}_2, .., \hat{v}_{N_3}]$ are two vectors both with N_3 evenly spaced elements between 0 and 1. Thus, the boundary loss for the neural copula is defined as

$$L^B(\boldsymbol{\theta}) = \frac{1}{4N_3}\Bigg(\sum_{i=1}^{N_3}|\tilde{C}([\hat{u}_i, 0];\boldsymbol{\theta})| + \sum_{i=1}^{N_3}|\tilde{C}([0, \hat{v}_i];\boldsymbol{\theta})| + \sum_{i=1}^{N_3}|\tilde{C}([\hat{u}_i, 1];\boldsymbol{\theta}) - \hat{u}_i| + \sum_{i=1}^{N_3}|\tilde{C}([1, \hat{v}_i];\boldsymbol{\theta}) - \hat{v}_i|\Bigg). \tag{7}$$

2) Nonnegativity Constraint of Neural Copula: The pdf values in the domain of copula pdf should be nonnegative. In the training process, we need to punish the case that the pdf values are negative. Formally, the nonnegativity constraint can be formulated as

$$\int_0^1\int_0^1 \mathrm{ReLU}(-f_{\tilde{C}}([\boldsymbol{u},\boldsymbol{v}];\boldsymbol{\theta}))d\boldsymbol{u}d\boldsymbol{v} = 0. \tag{8}$$

Similarly, suppose that $\dot{\boldsymbol{u}} = [\dot{u}_1, \dot{u}_2, \ldots, \dot{u}_{N_4}]$ and $\dot{\boldsymbol{v}} = [\dot{v}_1, \dot{v}_2, \ldots, \dot{v}_{N_4}]$ are two vectors both with N_4 evenly spaced elements between 0 and 1. The nonnegativity loss can be written as

$$L^N(\boldsymbol{\theta}) = \frac{1}{(N_4)^2}\sum_{i=1}^{N_4}\sum_{j=1}^{N_4}\mathrm{ReLU}\left(-f_{\tilde{C}}([\dot{u}_i, \dot{v}_j];\boldsymbol{\theta})\right) \tag{9}$$

where $\mathrm{ReLU}(\cdot)$ is the rectified linear unit function [60].

3) Integration Constraint of Neural Copula: The integration of the pdf values in the domain of copula pdf should be 1. Formally, the integration constraint can be formulated as

$$\int_0^1\int_0^1 f_{\tilde{C}}([\boldsymbol{u},\boldsymbol{v}];\boldsymbol{\theta})d\boldsymbol{u}d\boldsymbol{v} = 1. \tag{10}$$

The following steps in [33] are adopted to approximately implement the constraint of (10). First, to prevent the copula pdf from being negative, we need to conduct a nonnegative processing operation for the gradient $f_{\tilde{C}}([\boldsymbol{u},\boldsymbol{v}],\boldsymbol{\theta})$. Formally,

$$\bar{f}_{\tilde{C}}([\boldsymbol{u},\boldsymbol{v}];\boldsymbol{\theta}) = \mathrm{ReLU}(f_{\tilde{C}}([\boldsymbol{u},\boldsymbol{v}];\boldsymbol{\theta})) + \rho \tag{11}$$

where ρ is a minuscule positive value (set to 10^{-9} in our experiments). ρ is set to ensure that the input values of the log function in (13) are valid. Second, to implement the integral of pdfs corresponding to the discrete pixel values in remote sensing images, (10) is rewritten in an approximate form. Suppose that $\grave{\boldsymbol{u}} = [\grave{u}_1, \grave{u}_2, \ldots, \grave{u}_{N_5}]$ and $\grave{\boldsymbol{v}} = [\grave{v}_1, \grave{v}_2, \ldots, \grave{v}_{N_5}]$ are two vectors both with N_5 evenly spaced elements between 0 and 1, and $\Delta = 1/(N_5 - 1)$ is the space size. The integration loss from (10) can be approximated as

$$L^I(\boldsymbol{\theta}) = \left|1 - \sum_{i=1}^{N_5}\sum_{j=1}^{N_5}\bar{f}_{\tilde{C}}([\grave{u}_i, \grave{v}_j];\boldsymbol{\theta})\Delta^2\right|. \tag{12}$$

4) MLE for Neural Copula: To model the dependence relationships in heterogeneous image pairs, the pdf values by the neural copula model are required to be maximized in terms of $\tilde{\boldsymbol{u}}$ and $\tilde{\boldsymbol{v}}$. This is implemented by using maximum likelihood estimation (MLE), which has been widely used in the estimation for parameters in copula functions [35], [36]. The maximum likelihood loss is defined as follows:

$$L^M(\boldsymbol{\theta}) = \left|\eta - \frac{1}{N_2}\sum_{i=1}^{N_2}\log\left(\bar{f}_{\tilde{C}}([\tilde{u}_i, \tilde{v}_i];\boldsymbol{\theta})\right)\right|. \tag{13}$$

η should be a positive finite value in implementation. In our experiments, we have empirically found that the setting of $\eta = 10$ is satisfactory.

5) Observation Constraint of Neural Copula: To ensure consistency between the outputs of the neural copula and the empirical cdf of the original bi-temporal data (i.e., Γ^{T_1} and Γ^{T_2}), we need to sample some observation points and narrow difference between the outputs of the neural copula and the empirical cdf values at these points. Suppose that $\boldsymbol{x} = [x_1, x_2, \ldots, x_{N_6}]$ and $\boldsymbol{y} = [y_1, y_2, \ldots, y_{N_6}]$ are two vectors both with N_6 evenly spaced elements between 0 and 1. The observation points can be sampled as (x_i, y_j), where i and j here both range from 1 to N_6. The marginal cdf values of $\boldsymbol{x}$ and $\boldsymbol{y}$ generated from the distribution of Γ^{T_1} and Γ^{T_2} are $\bar{\boldsymbol{u}} = [\bar{u}_1, \bar{u}_2, \ldots, \bar{u}_{N_6}]$ and $\bar{\boldsymbol{v}} = [\bar{v}_1, \bar{v}_2, \ldots, \bar{v}_{N_6}]$, respectively,

$$\begin{aligned} \bar{\boldsymbol{u}} &= \mathrm{KDE}(\boldsymbol{x};\Gamma^{T_1}) \\ \bar{\boldsymbol{v}} &= \mathrm{KDE}(\boldsymbol{y};\Gamma^{T_2}) \end{aligned} \tag{14}$$

the observation loss can be written as

$$L^O(\boldsymbol{\theta}) = \frac{1}{(N_6)^2}\sum_{i=1}^{N_6}\sum_{j=1}^{N_6}\left|\tilde{C}([\bar{u}_i, \bar{v}_j];\boldsymbol{\theta}) - \frac{\sum_{m=1}^{N_2} I\left(\gamma_m^{T_1} \le x_i, \gamma_m^{T_2} \le y_j\right)}{N_2}\right| \tag{15}$$

where $I(\cdot)$ represents the indicator function

$$I(z) = \begin{cases} 1 & \text{if } z \text{ is true} \\ 0 & \text{if } z \text{ is not true.} \end{cases} \tag{16}$$

Finally, we combine the aforementioned five constraint loss terms to form the overall loss function,

$$L(\boldsymbol{\theta}) = w_B L^B(\boldsymbol{\theta}) + w_I L^I(\boldsymbol{\theta}) + w_N L^N(\boldsymbol{\theta}) + w_M L^M(\boldsymbol{\theta}) + w_O L^O(\boldsymbol{\theta}) \tag{17}$$

where we set the weights $[w_B, w_I, w_N, w_M, w_O]$ equaling $[2, 0.3, 1, 0.1, 5]$ in our experiments because they are empirically found to be robust in most scenarios.

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Algorithm 1 NN-Copula-CD

Input:

Bi-temporal heterogeneous images I^{T_1} and I^{T_2}.

Step 1: Superpixel Segmentation and Training Sample Selection

1) Segment heterogeneous image pair I^{T_1} and I^{T_2} into superpixel pairs Ω^{T_1} and Ω^{T_2}, respectively.
2) Select a set of unchanged superpixel pairs $\tilde{\Omega}^{T_1}$ and $\tilde{\Omega}^{T_2}$ from Ω^{T_1} and Ω^{T_2}, respectively, see (2).
3) Calculate the normalized superpixel intensity Γ^{T_1} and Γ^{T_2} of $\tilde{\Omega}^{T_1}$ and $\tilde{\Omega}^{T_2}$, respectively, see (3).

Step 2: Neural Copula Model Training

4) Estimate the marginal cdf values $\tilde{\boldsymbol{u}}$ and $\tilde{\boldsymbol{v}}$ with respect to Γ^{T_1} and Γ^{T_2}, respectively, see (4).
5) Train the neural copula model $\tilde{C}([\boldsymbol{u}, \boldsymbol{v}]; \boldsymbol{\theta})$ via loss functions designed according to copula theory, see (7), (9), (12), (13), and (15).

Step 3: Obtain CD Results

6) Similar to 1) and 3), segment heterogeneous image pair I^{T_1} and I^{T_2} into superpixel pairs Φ^{T_1} and Φ^{T_2}, , respectively. Then, calculate the normalized superpixel intensity Z^{T_1} and Z^{T_2} of Φ^{T_1} and Φ^{T_2}, respectively.
7) Estimate the marginal cdf values $\ddot{\boldsymbol{u}}$ and $\ddot{\boldsymbol{v}}$ with respect to Z^{T_1} and Z^{T_2}, respectively, see (18).
8) Input $\ddot{\boldsymbol{u}}$ and $\ddot{\boldsymbol{v}}$ to the trained neural copula model and then obtain degrees of dependence $\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta})$.
9) Apply fuzzy c-means clustering algorithm to classify $-\log_{10}(\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta}))$ into changed and unchanged categories.

Output:

Binary CD map.

D. Use of the Trained Model to Obtain CD Results

Once the training of the neural copula model has been completed, we utilize the trained model to infer changed areas between I^{T_1} and I^{T_2}. We refer to the use of the trained model to generate CD results as the testing stage. The first step of the testing stage is superpixel segmentation. Note that the superpixel segmentation results from the training stage can be directly used here to avoid redundant computation. However, during the testing stage, we once again apply the superpixel segmentation method in [13] to segment the original images I^{T_1} and I^{T_2} into superpixel sets, which may result in a different number of superpixels (denoted as N_7). The purpose of this repetition of superpixel segmentation is to ensure the superpixel scales in the training and testing stages are different so that the trained neural copula model does not show obvious over-fitting performance on a single superpixel scale. We will further discuss the impact of the number of superpixels on the CD performance in Section IV-E. The segmented superpixel sets of the original images I^{T_1} and I^{T_2} in the testing stage are denoted by Φ^{T_1} and Φ^{T_2}, respectively. Then, similar to (3), the normalized superpixel intensity values for superpixels in Φ^{T_1} and Φ^{T_2} are calculated and denoted by Z^{T_1} and Z^{T_2}, respectively.

The next step is to transform Z^{T_1} and Z^{T_2} into probability space, that is, to estimate the marginal cdf values for elements in Z^{T_1} and Z^{T_2}. We calculate the marginal cdf values of elements in Z^{T_1} and Z^{T_2} according to the marginal distributions fit by the training sets Γ^{T_1} and Γ^{T_2}, respectively,

$$\begin{aligned} \ddot{\boldsymbol{u}} &= \mathrm{KDE}\left(Z^{T_1}; \Gamma^{T_1}\right) \\ \ddot{\boldsymbol{v}} &= \mathrm{KDE}\left(Z^{T_2}; \Gamma^{T_2}\right) \end{aligned} \tag{18}$$

where $\ddot{\boldsymbol{u}} = [\ddot{u}_1, \ddot{u}_2, \ldots, \ddot{u}_{N_7}]$ and $\ddot{\boldsymbol{v}} = [\ddot{v}_1, \ddot{v}_2, \ldots, \ddot{v}_{N_7}]$ are the marginal cdf values of Z^{T_1} and Z^{T_2}, respectively. Then, we input $\ddot{\boldsymbol{u}}$ and $\ddot{\boldsymbol{v}}$ into the trained copula model and calculate the copula pdf values of the neural copula model via differentiation

$$\bar{f}_{\tilde{C}}\left([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta}\right) = \mathrm{ReLU}\left(\frac{\partial^2 \tilde{C}\left([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta}\right)}{\partial \ddot{\boldsymbol{u}} \partial \ddot{\boldsymbol{v}}}\right) + \rho \tag{19}$$

where $\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta})$ represents the degrees of dependence between superpixel pairs in bi-temporal images. An element in $\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta})$ with a smaller value means that the dependence of the corresponding superpixel pair is different from the dependence of those unchanged superpixel pairs utilized for training. Then, a negative logarithmic operation is applied on $\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta})$ to further enhance the discrimination between changed and unchanged categories (see Fig. 3). Afterward, we use the fuzzy c-means clustering algorithm [54] to divide $-\log_{10}(\bar{f}_{\tilde{C}}([\ddot{\boldsymbol{u}}, \ddot{\boldsymbol{v}}]; \boldsymbol{\theta}))$ into two categories, that is, classifying the superpixel pairs into changed and unchanged categories. Finally, we can obtain the CD results by assigning the categories of the superpixels to the corresponding pixels in heterogeneous bi-temporal images. We summarize the CD process of the proposed NN-Copula-CD in Algorithm 1.

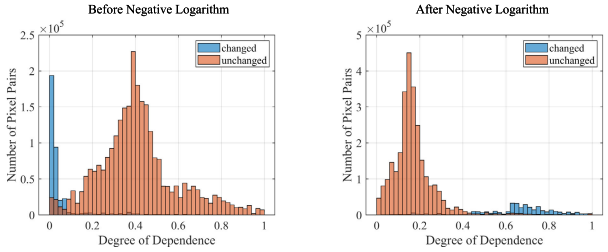


Fig. 3. Distribution of the degrees of dependence before and after a negative logarithmic operation. For clear illustration, the degrees of dependence have been min–max normalized.

IV. Experimental Results

In this section, we apply the proposed NN-Copula-CD algorithm on four heterogeneous datasets and compare its

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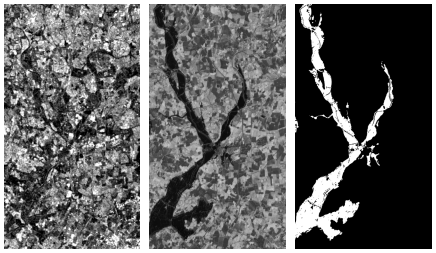


Fig. 4. First dataset. (a) Prechange SAR image was acquired in 1999. (b) Postchange NDVI image was acquired in 2000. (c) GT.

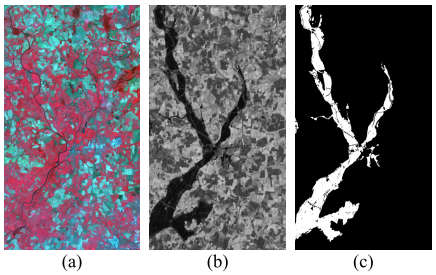


Fig. 5. Second dataset. (a) Prechange SPOT image was acquired in 1999. (b) Postchange NDVI image was acquired in 2000. (c) GT.

performance with that of several well-known heterogeneous CD methods. In addition, we conduct in-depth discussions and perturbation analysis to investigate the factors that influence the performance of our NN-Copula-CD algorithm.

In the proposed NN-Copula-CD method, it is necessary to preprocess input images over multiple channels into a single-channel form. The channel-dimension reduction preprocessing can significantly facilitate the efficiency of learning the major dependence between superpixel pairs with acceptable information loss. Consequently, multispectral images will undergo dimensionality reduction using the PCA algorithm [61] in our experiments.

A. Dataset

As shown in Fig. 4, the first dataset is of flood areas in Gloucester, England. It includes a prechange radar (SAR) image acquired by the European remote sensing satellite in 1999 and a postchange normalized difference vegetation index (NDVI) image acquired in 2000. NDVI images are obtained from the red (visible) band and the near-infrared band (which bands are captured by optical remote sensing devices [62]). The details of formulas for obtaining the NDVI image can be found in [63]. In Fig. 5, the second dataset is still of the flood areas in Gloucester, where the prechange multispectral image is acquired by the Satellite Pour l'Observation de la Terre (SPOT) satellite in 1999 and the postchange NDVI image is captured in 2000. The heterogeneous images in Figs. 4 and 5 are from [4], [19], and [23], and the size of the heterogeneous images in Figs. 4 and 5 is both 1318 × 2359. The postchange NDVI image used in the first dataset is the same as the one used in the second dataset. Following the strategy in [29], we adopt the same ground truth (GT) for the first and the second datasets.

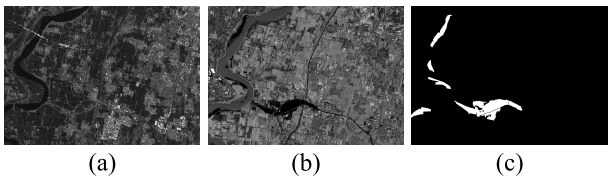


Fig. 6. Third dataset. (a) Prechange SAR image was acquired on May 11, 2021. (b) Postchange NIR image was acquired on October 11, 2021. (c) GT.

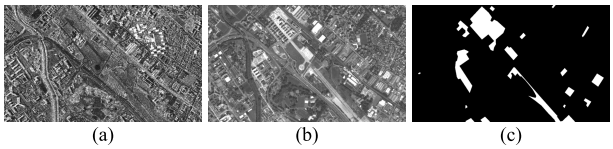


Fig. 7. Fourth dataset. (a) Prechange SAR image acquired in 2009. (b) Postchange optical image was acquired in 2013. (c) GT.

The third dataset displayed in Fig. 6 represents a flood event that occurred in Shandong Province, China. This dataset contains a prechange SAR image acquired by the Sentinel-1 satellite on May 11, 2021, and a postchange near-infrared (NIR) image acquired by the Sentinel-2 satellite on October 11, 2021. The prechange and postchange images have been registered and the sizes of the two images in this dataset are 1185 × 835. The GT is labeled based on the hydrological monitor data and the heterogeneous remote sensing data.

The fourth dataset in Fig. 7 is provided by [65] and it reflects the urban construction in Toulouse, from 2009 to 2013. This dataset is composed of a prechange SAR image acquired by the TerraSAR-X satellite in 2009 and a postchange optical image acquired by the Pleiades satellite in 2013. The optical image is originally provided in grayscale form with the size 4404 × 2600 and the size of the SAR image is 4404 × 2604. We crop the SAR image into the size 4404 × 2600, which is the same as the optical image.

B. Experimental Setting

1) Heterogeneous CD Algorithms for Performance Comparison: In our experiments, we compare the proposed NN-Copula-CD method with several representative methods[3] for heterogeneous CD.

Homogeneous pixel transformation (HPT) [19] method is a semi-supervised heterogeneous CD method based on HPT. It transforms one image into the space of the other image in both forward and backward forms, which aligns the pixel-level features of the two images in a homogeneous space. Then, the CD results are generated via the calculation of difference maps and the fuzzy c-means clustering algorithm.

Symmetric convolutional coupling network (SCCN) [20] is a well-known heterogeneous CD method based on the deep convolutional coupling network. First, SCCN exploits denoising autoencoders to initialize each side of the network. Then, the two sides of the network are collaboratively trained to transform the two images in the common deep feature

[3]We reimplemented HPT and SCCN based on the description provided in papers [19], [20], and reimplemented IRG-McS [13], IST-CRF [64], and VDF-HCD [56] using their public code.

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TABLE I
COMPARISON RESULTS OF DIFFERENT METHODS ON FOUR DATASETS. THE HIGHEST SCORE IS MARKED IN BOLD

	First Dataset PCC / IoU / **F1** / **KC** (%)	Second Dataset PCC / IoU / **F1** / **KC** (%)
IRG-McS [13]	93.33 / 60.04 / 75.03 / 71.24	94.59 / 65.03 / 78.81 / 75.73
SCCN [20]	95.22 / 63.47 / 77.65 / 75.02	94.57 / 62.77 / 77.13 / 74.05
HPT [19]	96.15 / 70.19 / 82.48 / 80.34	93.89 / 64.21 / 78.20 / 74.74
IST-CRF [64]	94.87 / 61.14 / 75.89 / 73.06	95.40 / 67.87 / 80.86 / 78.25
VDF-HCD [56]	94.01 / 53.11 / 69.38 / 66.23	94.93 / 65.87 / 79.43 / 76.54
NN-Copula-CD	**96.67 / 76.21 / 86.50 / 84.60**	**95.53 / 70.39 / 82.62 / 80.07**
	Third Dataset PCC / IoU / **F1** / **KC** (%)	**Fourth Dataset** PCC / IoU / **F1** / **KC** (%)
IRG-McS [13]	97.48 / 52.50 / 68.85 / 67.60	89.71 / 20.37 / 33.84 / 28.26
SCCN [20]	97.69 / 48.16 / 65.01 / 63.81	86.27 / 12.34 / 21.97 / 14.52
HPT [19]	97.14 / 48.13 / 64.98 / 63.56	**90.66** / 15.80 / 27.29 / 22.60
IST-CRF [64]	90.11 / 21.56 / 35.47 / 32.02	87.04 / 13.12 / 23.20 / 16.15
VDF-HCD [56]	94.13 / 29.13 / 45.11 / 42.53	81.98 / 18.27 / 30.90 / 22.31
NN-Copula-CD	**97.74 / 53.31 / 69.55 / 68.39**	84.99 / **25.38 / 40.49 / 33.22**

space. Finally, the CD results are obtained by classifying the difference map of the two sides of the network using a thresholding algorithm.

Iterative robust graph with Markovian co-segmentation (IRG-McS) method [13] is a heterogeneous CD method based on graph representation and Markov co-segmentation. The IRG-McS method first builds the KNN graph representations based on superpixels for bi-temporal heterogeneous images and then utilizes graph mapping and calculates the difference maps between graph representations. Finally, the CD results are generated from the difference maps based on a Markovian co-segmentation model.

Iterative structure transformation and conditional random field (IST-CRF) method [64] is a structure transformation-based method, leveraging graphs to represent similarity structure information of images. This method transforms bi-temporal images into the consistent differential domain to calculate difference maps and then utilizes a conditional random field segmentation model to recognize changed regions.

Vertex domain filtering-based heterogeneous CD (VDF-HCD) [56] converts the heterogeneous CD problem into a graph signal-processing problem, which constructs the relationship of the topological structures based on the KNN graphs of bi-temporal images. The vertex filtering operations on the graphs are leveraged in VDF-HCD to facilitate extracting the changes.

2) Performance Indicators: Four commonly used performance indicators are adopted in our experiments, that is, F1-score (F1), percentage of correct classification (PCC), intersection over union (IoU), and kappa coefficient (KC) to evaluate the performance of the proposed NN-Copula-CD method and competitors. The definitions of F1, PCC, IoU, and KC are as follows:

$$\begin{aligned} \mathrm{P} &= \mathrm{TP}\,/\,(\mathrm{TP}+\mathrm{FP}),\ \ \mathrm{R} = \mathrm{TP}\,/\,(\mathrm{TP}+\mathrm{FN}) \\ \mathrm{F1} &= 2\mathrm{PR}\,/\,(\mathrm{P}+\mathrm{R}) \\ \mathrm{PCC} &= (\mathrm{TP}+\mathrm{TN})\,/\,(\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}) \\ \mathrm{IoU} &= \mathrm{TP}\,/\,(\mathrm{TP}+\mathrm{FP}+\mathrm{FN}) \\ \mathrm{KC} &= (\mathrm{PCC}-\mathrm{PRE})\,/\,(1-\mathrm{PRE}) \\ \mathrm{PRE} &= \frac{(\mathrm{TP}+\mathrm{FN})(\mathrm{TP}+\mathrm{FP})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \end{aligned} \tag{20}$$

where TP, TN, FP, and FN denote the numbers of true positive, true negative, false positive, and false negative predictions, respectively. Note that the area of the changed region and the area of the unchanged region are usually unbalanced in CD. A higher PCC reflects better overall accuracy but often ignores the recall rate of the category with fewer quantities, while a higher KC or F1 indicates better comprehensive performance even if there is an imbalance between changed and unchanged areas. In practice, KC and F1 are often superior performance indicators than PCC when evaluating the comprehensive performance of a CD method.

3) Implementation Setting: Our NN-Copula-CD method is mainly implemented via the Keras framework [66] on the NVIDIA RTX TITAN GPU. In fact, the computational burden needed to train the neural copula model is small since it only contains five hidden layers. In the training process, we use the Nesterov Adam optimizer [67] to optimize the neural copula model, and the total training epochs are set to 25 000. When the training process is over, we choose the model with the least loss $L(\boldsymbol{\theta})$ as the testing model to produce the CD results. N_1 and N_7 are set to 6000 and 3000, respectively, for the first, second, and third datasets, while 8000 and 3000, respectively, for the fourth dataset. The increase in the number of superpixels for training is due to the larger image size in the fourth dataset. In this article, we set $N_3 = 400$, N_4, $N_5 = 256$, and $N_6 = 100$.

C. Performance Comparison

The quantitative performance of different methods on the four datasets is reported in Table I. The visual results of

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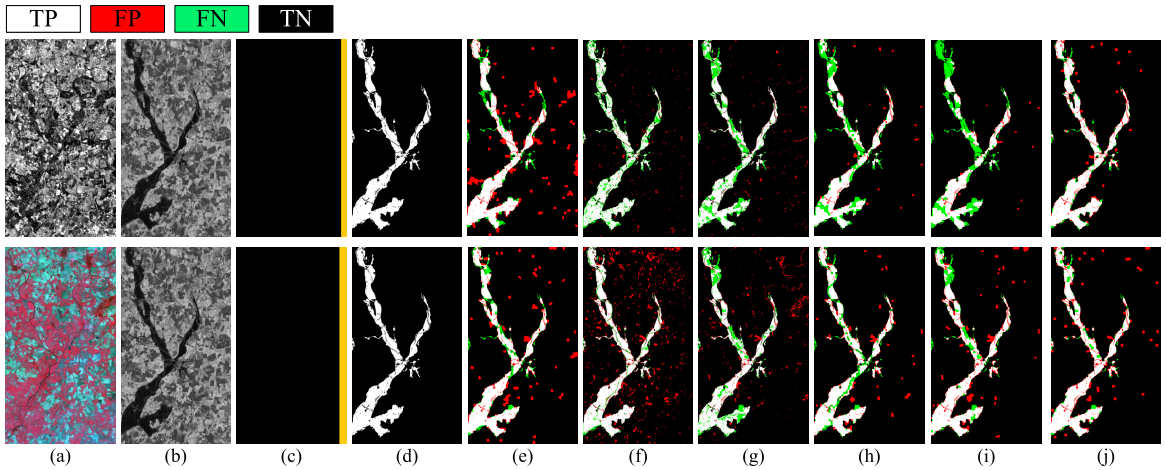


Fig. 8. Visualized CD results of different methods on the first and second datasets. (a) Prechange image. (b) Postchange image. (c) Selected region (marked by orange) is used to train the semi-supervised methods including HPT [19] and our NN-Copula-CD. (d) GT. (e) IRG-McS [13]. (f) HPT [19]. (g) SCCN [20]. (h) IST-CRF [64]. (i) VDF-HCD [56]. (j) Proposed NN-Copula-CD.

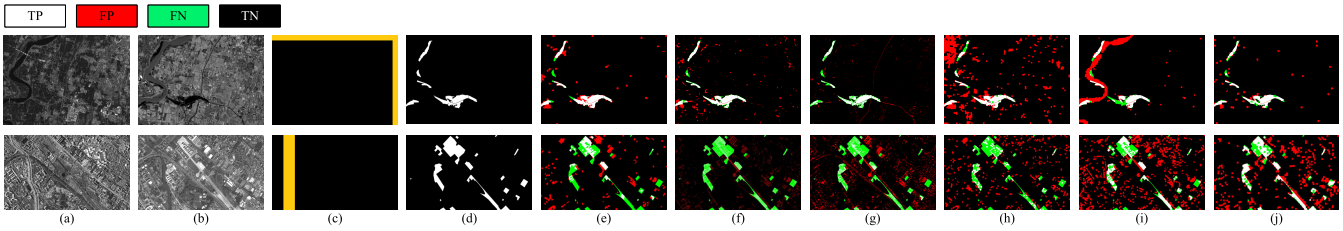


Fig. 9. Visualized CD maps of different methods on the third and fourth datasets. (a) Prechange image. (b) Postchange image. (c) Orange region is used to train HPT [19] and NN-Copula-CD. (d) GT. (e) IRG-McS [13]. (f) HPT [19]. (g) SCCN [20]. (h) IST-CRF [64]. (i) VDF-HCD [56]. (j) Our NN-Copula-CD.

different methods on the four datasets are shown in Figs. 8 and 9. Note that we exploit the same selected unchanged regions to train the semi-supervised methods including HPT [19] and our method, as shown in Figs. 8(c) and 9(c). The quantitative comparison in Table I shows that the proposed NN-Copula-CD algorithm achieves a higher score of F1 and KC on all four datasets. The semi-supervised competitor HPT [19] obtains good performance in the flood scenes, yet it exhibits a degraded performance in the city scene (i.e., the fourth dataset). The reason for this is the HPT [19] used in the city scene suffers from a much more complex heterogeneity than that in the flood scenes. Meanwhile, the serious speckle noise (especially on buildings) in SAR images also frequently degrades the performance of dense prediction tasks such as CD and semantic segmentation [68]. The DNN-based SCCN [20] also achieves good performance in flood scenes, yet it struggles to achieve robust detection in the city scene. The graph-based CD methods [13], [56], [64] mainly focus on constructing image structure dependence into graph representations, and comparing the transformed graph representations in the same domain via distance (e.g., L2 distance) to achieve CD. IRG-McS [13], IST-CRF [64], and VDF-HCD [56] achieve competitive CD performance across all the considered methods, while our NN-Copula-CD still achieves better CD performance than these graph-based methods on the evaluation indicators of KC and F1. This further demonstrates the effectiveness of the proposed NN-Copula-CD method. Compared with the existing representative competitors, the proposed NN-Copula-CD method significantly reduces the missed predictions, and the changed areas identified by the proposed NN-Copula-CD are more complete.

To interpret the results obtained by our NN-Copula-CD method intuitively and systematically, we visualize the mechanism of the whole CD process in Fig. 10. First, we draw the normalized superpixel intensities of the selected unchanged bi-temporal superpixels, that is, Γ^{T_1} and Γ^{T_2} (in the training process), as a 2-D scatter diagram, as shown in the second column of Fig. 10. These scatter pairs imply the unchanged patterns between bi-temporal images. Second, we train a neural network to learn the dependence between unchanged superpixels via copula theory. The third column of Fig. 10 represents the learned dependence represented by the joint pdf of normalized superpixel intensities of unchanged superpixel pairs. It can be clearly observed that the joint probability densities shown in the third column closely approximate the scatter distributions in the second column of Fig. 10. This demonstrates that the loss functions designed according to copula theory effectively guide the training of the neural network, which makes the trained neural network work as a copula function that is suitable for the given scenarios. In addition, visual comparisons between the second column and the third column of Fig. 10 also imply that the interpretability provided by copula theory can explain how the neural network works. Finally, we utilize the trained neural network to generate the degrees of dependence for all superpixel pairs

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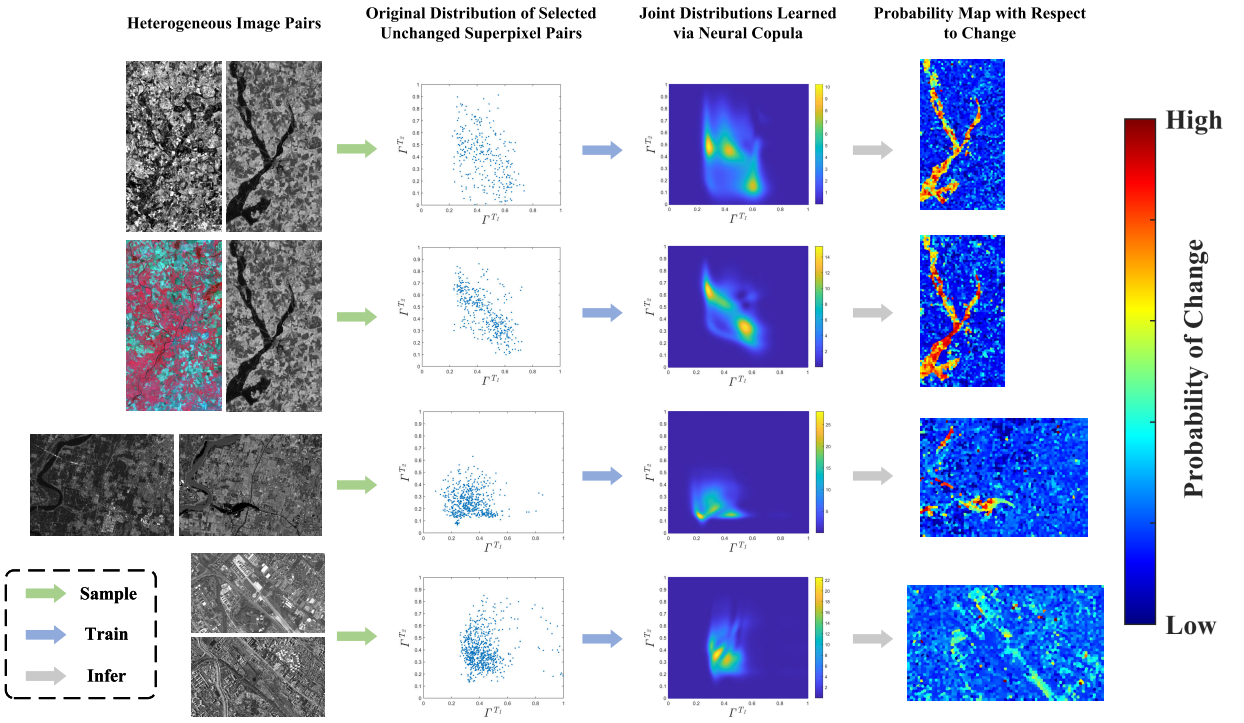


Fig. 10. Original distribution of selected unchanged training samples (i.e., Γ^{T_1} and Γ^{T_2}) and the simulated joint distributions learned by the neural copula. Based on the dependence of unchanged regions learned via the neural copula, NN-Copula-CD infers the probabilities with respect to the changed category for each superpixel pair in bi-temporal heterogeneous remote sensing images.

in bi-temporal images. Specifically, the cdf values of each superpixel pair are input into the trained neural network to infer the corresponding degrees of dependence, which can be further transformed and visualized in probability maps with respect to the changed category (as shown in the fourth column of Fig. 10). The inference process can be understood from a statistical perspective using the neural copula model, and then the changed and unchanged areas are easily distinguished based on the degrees of dependence. Compared with the existing DNN-based method [20], our NN-Copula-CD utilizes a simpler network structure (i.e., fully connected networks with five hidden layers) with better interpretability, which enables our method to maintain excellent performance using fewer network parameters instead of stacking more neural layers.

D. Discussion

1) Traditional Copula Functions Versus Neural Copula Model: To demonstrate the superiority of the neural copula compared with the traditional copula functions in the heterogeneous CD task, we replace the neural copula model with commonly used traditional copula functions, that is, Student-*t* copula, Gaussian copula, Clayton copula, and Frank copula. Note that the reference [29] manually chooses the best traditional copula from various traditional copulas via the visual observation of dependence generated from a graphical method [69]. However, the dependence is usually different for various scenes, where the best traditional copula for various scenes is also different and should be individually selected. In our NN-Copula-CD, the neural network provides more parameters, that is, degrees of freedom, than the commonly used traditional copulas, when manually searching for the best copula function from existing traditional copula functions is replaced by learning a copula function that is better suited for the given scenes. The quantitative results in Table II show that the neural copula surpasses the commonly used traditional copulas on four datasets. This is because the neural copula model learns the complex dependence from heterogeneous remote sensing data more accurately.

TABLE II
COMPARISON RESULTS BETWEEN NEURAL COPULA IN OUR NN-COPULA-CD AND TRADITIONAL COPULAS ON FOUR DATASETS. THE HIGHEST SCORE IS MARKED IN BOLD

	First Dataset **F1 / KC** (%)	Second Dataset **F1 / KC** (%)
Gaussian Copula	71.79 / 68.92	81.05 / 78.78
Student-*t* Copula	73.07 / 70.25	82.26 / 79.53
Clayton Copula	52.38 / 49.03	54.81 / 51.39
Frank Copula	63.16 / 57.60	70.90 / 66.25
Neural Copula	**86.50 / 84.60**	**82.62 / 80.07**
	Third Dataset **F1 / KC** (%)	Fourth Dataset **F1 / KC** (%)
Gaussian Copula	19.97 / 15.35	14.55 / 2.07
Student-*t* Copula	25.11 / 20.78	37.83 / 30.22
Clayton Copula	6.98 / 1.01	16.41 / 2.40
Frank Copula	11.75 / 6.24	17.02 / 4.12
Neural Copula	**69.55 / 68.39**	**40.49 / 33.22**

To more intuitively understand the superiority of the neural copula over the traditional copulas, we visualize the joint

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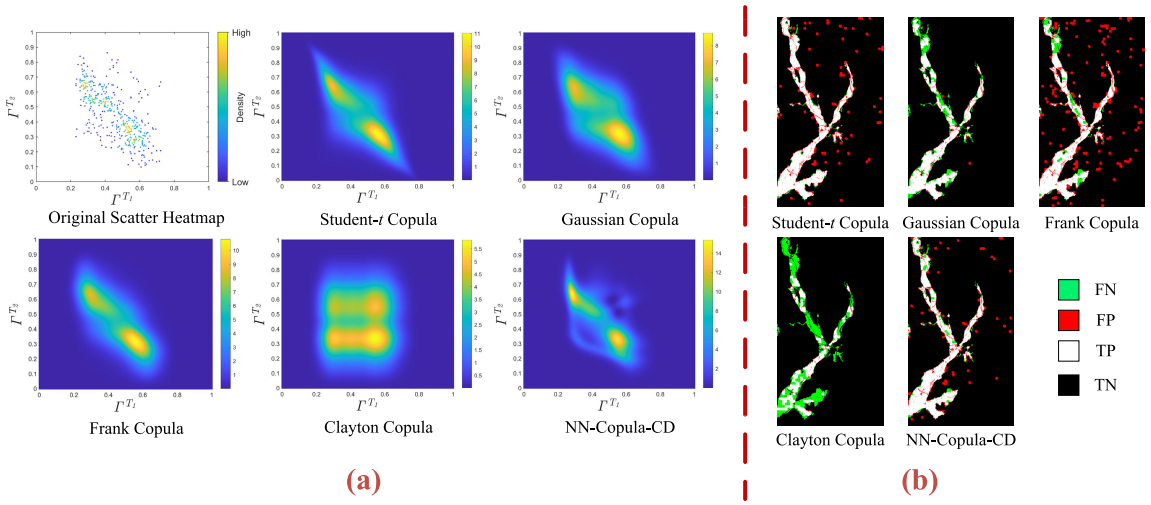


Fig. 11. (a) Joint distributions of Γ^{T_1} and Γ^{T_2} fit using NN-Copula-CD and traditional copula functions, respectively. (b) Change maps using NN-Copula-CD and traditional copula functions, respectively.

distribution of Γ^{T_1} and Γ^{T_2} fit by neural copula and traditional copulas on the second dataset in Fig. 11(a). Note that the traditional copulas are defined in terms of fixed mathematical models and tend to fit the data in a more rigid fashion resulting in under-fitting. Meanwhile, the powerful nonlinear modeling capabilities of neural networks provide more flexible mathematical models, which help our NN-Copula-CD to learn the dependence and approach the distribution of original data more closely. In addition, the visual comparisons of the change maps between traditional copula functions and NN-Copula-CD in Fig. 11(b) demonstrate that NN-Copula-CD surpasses the CD performance of traditional copulas when the neural copula model adheres to the mathematical constraints of copula theory. Therefore, neural copula is more capable than traditional copulas to capture the complex dependence in the above heterogeneous remote sensing scenes.

2) Comparison Between DNN-Based Method and Our NN-Copula-CD Under the Same Training Data for Semi-Supervised CD: To fairly compare our NN-Copula-CD and the DNN-based SCCN [20] using the same training data, we use the same unchanged training regions in our NN-Copula-CD to supervise the training of SCCN [20]. Inspired by the experiments conducted in [20] implementing the supervised version of SCCN [20], we implement a semi-supervised version of SCCN [20]. During the updating process of the probability map[4] (generated by SCCN [20]) that represents the likelihood of each pixel of unchanged category, we fix the probabilities of unchanged category in the training regions selected by us at 1. Thus, the deep convolutional coupling network supervised by the probability map can employ the same supervised information as our NN-Copula-CD in every backpropagation. Table III displays the comparison results of our NN-Copula-CD and the semi-supervised SCCN. The quantitative results demonstrate that our NN-Copula-CD still achieves superior performance compared with the semi-supervised SCCN in terms of KC and F1 under the same supervised information on both city and flood scenes.

[4]The definition and details about the probability map can be found in Section III-C of [20].

TABLE III

COMPARISON RESULTS BETWEEN NN-COPULA-CD AND SEMI-SUPERVISED SCCN. THE HIGHEST SCORE IS MARKED IN BOLD

	NN-Copula-CD **F1 / KC** (%)	Semi-supervised SCCN **F1 / KC** (%)
First Dataset	**86.50 / 84.60**	84.44 / 82.28
Fourth Dataset	**40.49 / 33.22**	32.64 / 25.74

TABLE IV

QUANTITATIVE EVALUATION WITH DIFFERENT NUMBERS OF TRAINING SAMPLES. THE PERFORMANCE INDICATORS ARE WRITTEN IN (%)

Number of Training Samples	**F1**	**KC**
5.31%	86.50	84.60
9.94%	84.68	82.58
16.62%	82.38	80.03

3) Effect of Training Sample Sizes: To investigate the impact of different numbers of training samples on the performance of our NN-Copula-CD algorithm, we conduct the corresponding experiments on the first dataset. We use three training regions containing 5.3%, 9.9%, and 16.6% pixels of the original images (as shown in Fig. 12), respectively, to train the neural copula model with other settings remaining fixed. The quantitative results using different numbers of training samples are shown in Table IV. The visual results corresponding to different numbers of training samples are displayed in Fig. 12. We can observe that the performance of NN-Copula-CD achieves the highest KC of 84.6% using 5.3% training samples. With more training samples, the performance of the NN-Copula-CD method does not show an expected improvement. We speculate that two factors may be contributing to this phenomenon. The first factor is that increasing the number of training samples brings higher modeling challenges,

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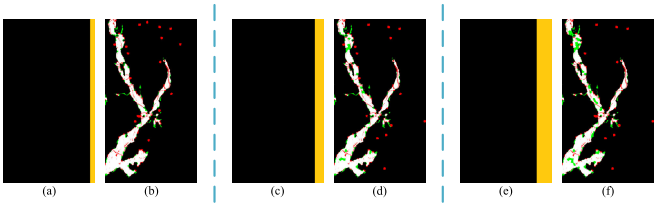


Fig. 12. CD results using different numbers of training samples. (a) Training samples with 5.31% pixels. (b) CD result of our method using 5.31% training samples. (c) Training samples with 9.94% pixels. (d) CD result of our method with 9.94% training samples. (e) Training samples with 16.62% pixels. (f) CD result of our method using 16.62% training samples. The FP, FN, TP, and TN cases are labeled in red, green, white, and black, respectively.

TABLE V

QUANTITATIVE RESULTS WITH DIFFERENT NUMBERS OF PARAMETERS IN NN-COPULA-CD. THE PERFORMANCE INDICATORS ARE REPRESENTED IN (%)

Model Size	F1	KC
Small (481 Params)	85.26	83.27
Medium (1021 Params)	86.73	84.86
Large (1761 Params)	86.50	84.60

and the concise neural network structure may not adapt to the growing number of samples. The second factor is that simply increasing the number of training samples is not always effective for enhancing the CD performance. Although our process to increase the number of training samples by expanding the width of rigid training regions is time-efficient and labor-saving, it may bring more low-quality training samples that degrade the CD performance. Note that the degradation of NN-Copula-CD is tolerable and its KC values are still higher than most competitive methods in Table I, which also verifies the robustness of our approach.

4) Effect of Number of Trainable Parameters in NN-Copula-CD: To investigate the effect of the number of trainable parameters in NN-Copula-CD, we have developed three variations of NN-Copula-CD, that is, the small version with 481 parameters, the medium version with 1021 parameters, and the large version with 1761 parameters. The small, medium, and large models consist of five hidden layers with 10, 15, and 20 neurons in each layer, respectively. We test the three variations of NN-Copula-CD with 5.3% training samples on the first dataset. As the quantitative comparison shown in Table V indicates, the medium version surpasses the other two versions on two indicators. Specifically, the small version and the large version show slight deviations in KC compared with the medium version.

5) Efficiency of NN-Copula-CD: We utilize run times to quantitatively measure the efficiency of our NN-Copula-CD. Table VI reports the efficiency of NN-Copula-CD and other competitors on two datasets with different image sizes. It can be observed that the VDF-HCD method [56] has higher efficiency among all compared methods, yet the CD performance of VDF-HCD is noticeably lower than our NN-Copula-CD on the first and fourth datasets. In the first dataset with a relatively small image size, the run time of IST-CRF[5] [64] is lower than that of our NN-Copula-CD. When the image size increases significantly in the fourth dataset, the run time of our NN-Copula-CD is lower than IST-CRF. Also, our NN-Copula-CD has better CD performance in all four datasets compared with IST-CRF. Note that the proposed NN-Copula-CD method achieves superior computational performance than the other training-based methods HPT [19] and SCCN [20]. This is because the enhanced interpretability facilitates our NN-Copula-CD working via a concise fully connected neural network instead of DNNs, leading to a reduction of computational complexity. Moreover, the concise design of the neural network in NN-Copula-CD also reduces the trainable parameters where the computation burdens on hardware are alleviated.

[5]In experiments, run time of IST-CRF contains the time of grid search for optimizing parameters α and β that are experimented in the article [64].

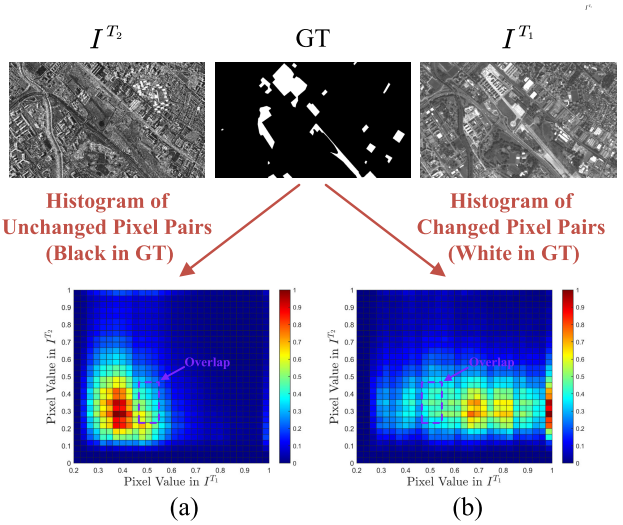


Fig. 13. Histograms of bi-temporal pixel values in unchanged/changed [i.e., (a)/(b)] regions according to the GT of the fourth dataset. The higher/lower histogram statistics are represented by the redder/bluer color. The higher histogram statistics denote that there are more pixel pairs distributed with the corresponding pixel values. The region marked by the purple line shows that there is an overlap of histogram statistics of changed and unchanged pixel pairs. This means that both changed and unchanged pixel pairs in the fourth dataset have the likelihoods of occurrence with the pixel values in the region marked by the purple line.

6) Discussion of FP Cases on City Scenario: We believe that the higher FP cases on the fourth dataset are highly related to the intrinsically high complexity of the city scenario of the fourth dataset. This city scenario is more complex than the flood scenarios, especially for SAR images that contain serious speckle noise corresponding to various buildings. Due to this speckle noise, the same buildings often have significantly different characteristics compared with those in optical images. Besides, the changed categories of land-cover objects in this city scenario are more diverse than other scenarios.

To show the complexity of the fourth dataset more intuitively, we provide the histograms of the pixel values of two heterogeneous images of the fourth dataset in Fig. 13. The pixel values in two images are normalized into [0, 1] via a division by 255. The histogram statistics are min–max normalized into [0, 1]. The redder/bluer color in Fig. 13 represents that there are more/less pixel pairs distributed with the corresponding pixel values in the histograms. According to the unchanged regions and changed regions labeled in

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TABLE VI
EFFICIENCY OF DIFFERENT METHODS

Dataset	Image Size	Run Time (s)				
		HPT† [19]	SCCN† [20]	IST-CRF‡ [64]	VDF-HCD‡ [56]	NN-Copula-CD†
First Dataset	1318×2359	3403	17669	656	27	893
Fourth Dataset	4404×2600	23007	68417	2037	45	1227

† denotes the implementations in Python, ‡ denotes the implementations in MATLAB

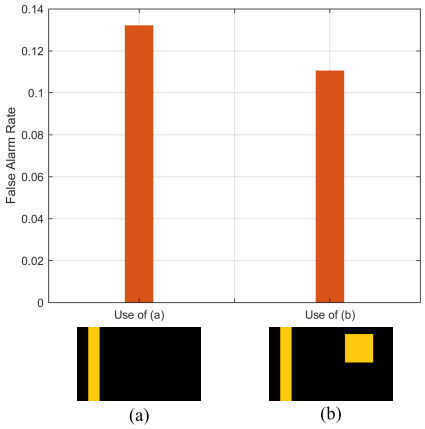


Fig. 14. False alarm rate with different training samples. (a) Before including additional training samples. (b) After including additional training samples. The training samples are marked orange.

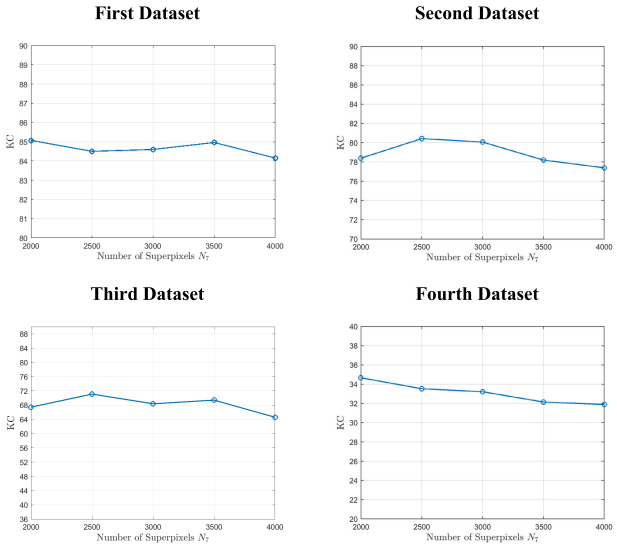


Fig. 15. Perturbation analysis regarding the number of superpixels (N_7).

GT, respectively, we show the histograms of the pixel values separately in Fig. 13(a) (corresponding to the bi-temporal pixels in unchanged regions) and Fig. 13(b) (corresponding to the bi-temporal pixels in changed regions). Fig. 13 shows that although the histograms of changed and unchanged pixel pairs have distinct distribution in most cases, there is an overlap of histogram statistics of changed and unchanged pixel pairs in the purple box. This overlap means that a large number of changed and unchanged pixel pairs in the fourth dataset are originally distributed with similar pixel values in the purple box. Thus, it is intrinsically complex in statistics to determine a clear classification boundary of changed and unchanged categories for the pixel pairs in the purple box, leading to some FPs in the CD results.

To reduce FP cases corresponding to the fourth dataset, we conduct a comparative experiment by adding more training samples for our NN-Copula-CD. The original training samples on the fourth dataset corresponding to Table I are shown in Fig. 14(a). Considering that buildings are a major component of the city, we add a square area that contains a dense cluster of buildings as new training samples to train our NN-Copula-CD, as shown in Fig. 14(b). Here, we use the false alarm (FA) rate to evaluate the number of FP cases in the CD results. FA rate is represented by FP/(TN+FP), where a higher FA rate indicates more FP cases in the CD results. Compared with the CD result corresponding to the training samples in Fig. 14(a), the quantitative CD result of using more training samples in Fig. 14(b) shows lower FA rate. This demonstrates that the high-quality manually selected training samples can effectively improve the performance of the proposed NN-Copula-CD on FP cases.

TABLE VII
PERTURBATION ANALYSIS ON THE LOSS WEIGHTS OF DIFFERENT MATHEMATICAL CONSTRAINTS IN (17) USED FOR COPULA GENERATION

	F1 / KC (%)		**F1 / KC** (%)
Baseline	86.50 / 84.60	w_B=1.5 w_B=2.5	86.62 / 84.75 86.03 / 84.06
w_N=0.75 w_N=1.25	85.35 / 83.34 85.90 / 83.91	w_I=0.225 w_I=0.375	86.58 / 84.69 86.13 / 84.16
w_M=0.075 w_M=0.125	84.72 / 82.62 86.42 / 84.51	w_O=3.75 w_O=6.25	85.08 / 83.03 85.96 / 83.99

E. Perturbation Analysis

To investigate the sensitivity of our model to the hyperparameters, we carry on the perturbation analysis for N_7 (i.e., the number of superpixels used for testing) and the loss weights $[w_B, w_I, w_N, w_M, w_O]$ for neural copula generation in the training stage.

1) Number of Superpixels: In Sections IV-C and IV-D, we set the number of superpixels to 3000 (i.e., $N_7 = 3000$) as a baseline to conduct the comparative experiments and discussions. Here, we fix other experimental conditions and apply a perturbation on the number of superpixels N_7, to test NN-Copula-CD, as illustrated in Fig. 15. It can be observed that compared with the baseline, KC with different numbers of superpixels is within moderate deviation on four datasets. This demonstrates the robustness of our NN-Copula-CD to the number of superpixels during model testing.

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2) Weights in the Loss Function for Neural Copula Generation: The training loss for our NN-Copula-CD in (17) is composed of five mathematical constraints and we use a set of empirical weights in experiments. To show that the selected empirical weights are not the only values for the neural copula generation, we conduct a perturbation analysis for the weight of each mathematical constraint on the first dataset. Specifically, we change one weight at a time while keeping others fixed. As shown in Table VII, the performance of our method is still relatively consistent when each weight in $[w_B, w_I, w_N, w_M, w_O]$ changes within a 25% range of the baseline value (i.e., $[w_B, w_I, w_N, w_M, w_O] = [2,0.3,1,0.1,5]$).

V. Conclusion

In this article, we propose a novel semi-supervised method based on the copula-guided neural network, named NN-Copula-CD, to address the CD problem in heterogeneous remote sensing images. In the proposed NN-Copula-CD, we treat CD as an inference problem in terms of statistical dependence, where the changes can be efficiently identified by comparing the degrees of dependence of remote sensing region pairs. In NN-Copula-CD, the neural network that is trained to measure the statistical dependence in bi-temporal images satisfies the characteristics of a copula function under the framework of copula theory. Therefore, our NN-Copula-CD method is endowed with in-model interpretability in terms of the CD mechanism using neural networks. In addition, our NN-Copula-CD is more robust in modeling the dependence in heterogeneous remote sensing datasets than traditional copula functions due to the powerful representation capabilities of neural networks. The in-depth experiments on four heterogeneous CD datasets demonstrate the effectiveness of our NN-Copula-CD method in comparison with the existing advanced heterogeneous CD methods, including the DNN-based SCCN. In the future, we will investigate the improvement of our NN-Copula-CD method in complex city scenarios.

References

[1] W. Li, X. Wang, G. Li, B. Geng, and P. K. Varshney, "Change detection in heterogeneous remote sensing images based on copula-supervised neural networks," in *Proc. IEEE Int. Conf. Signal, Inf. Data Process.*, Nov. 2024, in press.

[2] A. Asokan and J. Anitha, "Change detection techniques for remote sensing applications: A survey," *Earth Sci. Inform.*, vol. 12, no. 2, pp. 143–160, Jun. 2019.

[3] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.

[4] X. Jiang, G. Li, X.-P. Zhang, and Y. He, "A semisupervised Siamese network for efficient change detection in heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4700718.

[5] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[6] M. K. Ridd and J. Liu, "A comparison of four algorithms for change detection in an urban environment," *Remote Sens. Environ.*, vol. 63, no. 2, pp. 95–100, Feb. 1998.

[7] C. Benedek, X. Descombes, and J. Zerubia, "Building development monitoring in multitemporal remotely sensed image pairs with stochastic birth-death dynamics," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 1, pp. 33–50, Jan. 2012.

[8] S. H. Khan, X. He, F. Porikli, and M. Bennamoun, "Forest change detection in incomplete satellite images with deep neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5407–5423, Sep. 2017.

[9] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.

[10] Z. Lv et al., "Land cover change detection with heterogeneous remote sensing images: Review, progress, and perspective," *Proc. IEEE*, vol. 110, no. 12, pp. 1976–1991, Dec. 2022.

[11] H. Chen, N. Yokoya, C. Wu, and B. Du, "Unsupervised multimodal change detection based on structural relationship graph representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Dec. 2022, Art. no. 5635318.

[12] Y. Sun, L. Lei, D. Guan, M. Li, and G. Kuang, "Sparse-constrained adaptive structure consistency-based unsupervised image regression for heterogeneous remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4405814.

[13] Y. Sun, L. Lei, D. Guan, and G. Kuang, "Iterative robust graph for unsupervised change detection of heterogeneous remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 6277–6291, 2021.

[14] P. J. Howarth and G. M. Wickware, "Procedures for change detection using Landsat digital data," *Int. J. Remote Sens.*, vol. 2, no. 3, pp. 277–291, Jul. 1981.

[15] R. Jackson, "Spectral indices in N-space," *Remote Sens. Environ.*, vol. 13, no. 5, pp. 409–421, Nov. 1983.

[16] W. J. Todd, "Urban and regional land use change detected by using Landsat data," *J. Res. U.S. Geological Surv.*, vol. 5, no. 5, pp. 529–534, 1977.

[17] L. Wan, Y. Xiang, and H. You, "A post-classification comparison method for SAR and optical images change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1026–1030, Jul. 2019.

[18] D. Brunner, G. Lemoine, and L. Bruzzone, "Earthquake damage assessment of buildings using VHR optical and SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 5, pp. 2403–2420, May 2010.

[19] Z. Liu, G. Li, G. Mercier, Y. He, and Q. Pan, "Change detection in heterogenous remote sensing images via homogeneous pixel transformation," *IEEE Trans. Image Process.*, vol. 27, no. 4, pp. 1822–1834, Apr. 2018.

[20] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Mar. 2018.

[21] X. Jiang, G. Li, Y. Liu, X.-P. Zhang, and Y. He, "Change detection in heterogeneous optical and SAR remote sensing images via deep homogeneous feature fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 1551–1566, 2020.

[22] X. Li, Z. Du, Y. Huang, and Z. Tan, "A deep translation (GAN) based change detection network for optical and SAR remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 179, pp. 14–34, Sep. 2021.

[23] Z.-G. Liu, Z.-W. Zhang, Q. Pan, and L.-B. Ning, "Unsupervised change detection from heterogeneous data based on image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403413.

[24] X. Niu, M. Gong, T. Zhan, and Y. Yang, "A conditional adversarial network for change detection in heterogeneous images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 45–49, Jan. 2018.

[25] L. T. Luppino et al., "Deep image translation with an affinity-based change prior for unsupervised multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4700422.

[26] L. T. Luppino et al., "Code-aligned autoencoders for unsupervised change detection in multimodal remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 60–72, Jan. 2022.

[27] T. Schmidt, "Coping with copulas," in *Copulas-From Theory To Application in Finance*. London, U.K.: Risk Books, 2007, pp. 3–34.

[28] C. Li, G. Li, X. Wang, and P. K. Varshney, "A copula-based method for change detection with multisensor optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5620015.

[29] G. Mercier, G. Moser, and S. B. Serpico, "Conditional copulas for change detection in heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1428–1441, May 2008.

[30] U. Cherubini, E. Luciano, and W. Vecchiato, *Copula Methods in Finance*. Hoboken, NJ, USA: Wiley, 2004.

[31] T. Janke, M. Ghanmi, and F. Steinke, "Implicit generative copulas," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 26028–26039.

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[32] Y. Ng, A. A. Hasan, K. Elkhalil, and V. Tarokh, "Generative Archimedean copulas," in *Proc. 37th Conf. Uncertainty Artif. Intell.*, Jul. 2021, pp. 643–653.

[33] Z. Zeng and T. Wang, "Neural copula: A unified framework for estimating generic high-dimensional copula functions," 2022, *arXiv:2205.15031*.

[34] M. Sklar, "Fonctions de répartition à n dimensions et leurs marges," Publ. Inst. statist. Univ. Paris, Paris, France, 1959, vol. 8, pp. 229–231.

[35] S. G. Iyengar, P. K. Varshney, and T. Damarla, "A parametric copula-based framework for hypothesis testing using heterogeneous data," *IEEE Trans. Signal Process.*, vol. 59, no. 5, pp. 2308–2319, May 2011.

[36] S. Zhang, L. N. Theagarajan, S. Choi, and P. K. Varshney, "Fusion of correlated decisions using regular vine copulas," *IEEE Trans. Signal Process.*, vol. 67, no. 8, pp. 2066–2079, Apr. 2019.

[37] X. Wang, D. Zhu, G. Li, X.-P. Zhang, and Y. He, "Proposal-copula-based fusion of spaceborne and airborne SAR images for ship target detection," *Inf. Fusion*, vol. 77, pp. 247–260, Jan. 2022.

[38] C. Rudin, C. Chen, Z. Chen, H. Huang, L. Semenova, and C. Zhong, "Interpretable machine learning: Fundamental principles and 10 grand challenges," *Statist. Surveys*, vol. 16, pp. 1–85, Jan. 2022.

[39] C. Xu, Z. Liao, C. Li, X. Zhou, and R. Xie, "Review on interpretable machine learning in smart grid," *Energies*, vol. 15, no. 12, p. 4427, Jun. 2022.

[40] Y. Zhang, P. Tino, A. Leonardis, and K. Tang, "A survey on neural network interpretability," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 5, no. 5, pp. 726–742, Oct. 2021.

[41] D. V. Carvalho, E. M. Pereira, and J. S. Cardoso, "Machine learning interpretability: A survey on methods and metrics," *Electronics*, vol. 8, no. 8, p. 832, 2019.

[42] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, Feb. 2019.

[43] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[44] A. Guo, R. Dian, and S. Li, "A deep framework for hyperspectral image fusion between different satellites," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 7939–7954, Jul. 2023.

[45] Z. Kuang, H. Bi, F. Li, C. Xu, and J. Sun, "Polarimetry-inspired contrastive learning for class-imbalanced PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5212819.

[46] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[47] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[48] H. Chen et al., "Continuous cross-resolution remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623320.

[49] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[50] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[51] Y. Pang, J. Lin, T. Qin, and Z. Chen, "Image-to-image translation: Methods and applications," *IEEE Trans. Multimedia*, vol. 24, pp. 3859–3881, 2022.

[52] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[53] J. Liu, W. Zhang, F. Liu, and L. Xiao, "A probabilistic model based on bipartite convolutional neural network for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4701514.

[54] J. C. Bezdek, R. Ehrlich, and W. Full, "FCM: The fuzzy c-means clustering algorithm," *Comput. Geosci.*, vol. 10, nos. 2–3, pp. 191–203, Jan. 1984.

[55] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[56] Y. Sun, L. Lei, D. Guan, G. Kuang, and L. Liu, "Graph signal processing for heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4415823.

[57] S. Liu and M. E. Hodgson, "Satellite image collection modeling for large area hazard emergency response," *ISPRS J. Photogramm. Remote Sens.*, vol. 118, pp. 13–21, Aug. 2016.

[58] O. Ozdemir, T. G. Allen, S. Choi, T. Wimalajeewa, and P. K. Varshney, "Copula based classifier fusion under statistical dependence," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 11, pp. 2740–2748, Nov. 2018.

[59] E. Parzen, "On estimation of a probability density function and mode," *Ann. Math. Statist.*, vol. 33, no. 3, pp. 1065–1076, Sep. 1962.

[60] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[61] H. Abdi and L. J. Williams, "Principal component analysis," *WIREs Comput. Statistic*, vol. 2, no. 4, pp. 433–459, Jul. 2010.

[62] G. P. O. Reddy, "Satellite remote sensing sensors: Principles and applications," in *Geospatial Technologies in Land Resources Mapping, Monitoring and Management*. Cham, Switzerland: Springer, 2018, pp. 21–43.

[63] H. Q. Liu and A. Huete, "A feedback based modification of the NDVI to minimize canopy background and atmospheric noise," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 2, pp. 457–465, Mar. 1995.

[64] Y. Sun, L. Lei, D. Guan, J. Wu, and G. Kuang, "Iterative structure transformation and conditional random field based method for unsupervised multimodal change detection," *Pattern Recognit.*, vol. 131, Nov. 2022, Art. no. 108845.

[65] R. Touati, M. Mignotte, and M. Dahmane, "Multimodal change detection in remote sensing images using an unsupervised pixel pairwise-based Markov random field model," *IEEE Trans. Image Process.*, vol. 29, pp. 757–767, 2020.

[66] A. Gulli and S. Pal, *Deep Learning With Keras*. Birmingham, U.K.: Packt Publishing, 2017.

[67] T. Dozat, "Incorporating Nesterov momentum into Adam," in *Proc. 4th Int. Conf. Learn. Represent.*, 2016, pp. 1–4.

[68] H. Bi, L. Xu, X. Cao, Y. Xue, and Z. Xu, "Polarimetric SAR image semantic segmentation with 3D discrete wavelet transform and Markov random field," *IEEE Trans. Image Process.*, vol. 29, pp. 6601–6614, 2020.

[69] C. Genest and J.-C. Boies, "Detecting dependence with Kendall plots," *Amer. Statistician*, vol. 57, no. 4, pp. 275–284, Nov. 2003.

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